SAR Image Object Detection Network Based on Wavelet-Domain Soft-Thresholding and Frequency Attention

Haosheng Song, Weidi Xu, Jinyong Chen, Huipan Zhao, Lin Wang, and Hua Yu

***Abstract*—Ship detection in synthetic aperture radar (SAR) images is constrained by the dual challenges of speckle noise interference and multiscale variations of targets. Traditional convolutional neural networks (CNNs) struggle to effectively decouple high-frequency target details from noise because they process features solely in the spatial domain, leading to limited detection accuracy in complex scenes. To address this issue, this letter proposes an object detection method based on an SAR adaptive frequency attention mechanism SAFA-WT. This method integrates a learnable soft-thresholding (ST) module and a frequency attention (FA) mechanism within the wavelet transform (WT) domain. The former physically suppresses high-frequency clutter, while the latter dynamically weights multifrequency features to jointly optimize denoising and feature refinement. Experiments on the SAR ship detection dataset (SSDD) and high-resolution SAR images dataset (HRSID) benchmark datasets demonstrate that SAFANet exhibits significant robustness in strongly interfered scenes, outperforming mainstream algorithms such as YOLO26-OBB. Ablation studies further verify the effectiveness of SAFA-WT as a plug-and-play module. The related code is available at https://github.com/DRAFT-BEER/SAFA-WT**

***Index Terms*—Frequency attention (FA), rotated object detection, ship detection and classification, synthetic aperture radar (SAR), wavelet transform (WT).**

I. Introduction

SHIP detection in high-resolution synthetic aperture radar (SAR) images is critical for maritime monitoring but is severely hindered by inherent speckle noise and multiscale target variations. Particularly in high sea-state scenarios, the strong interference of speckle noise and the multiscale variations of ship targets severely constrain the detection performance of traditional algorithms, as shown in Fig. 1.

While deep learning detectors like Faster R-CNN [1], RetinaNet [2], and YOLO [3] series excel in general vision, transferring them to SAR images faces a fundamental obstacle: inherent speckle noise caused by coherent imaging. This noise degrades image quality and blurs boundaries, creating a unique "high-frequency paradox" during feature extraction: high-value fine-grained target details (e.g., ship edges) and destructive speckle noise both concentrate primarily in the high-frequency signal components. Traditional spatial-domain convolutional neural networks (CNNs) struggle to effectively decouple these two sources. They often misjudge high-frequency noise as valid features, elevating the false alarm rate (FAR) in complex clutter backgrounds. Conversely, employing aggressive downsampling or spatial filtering to suppress this noise inevitably leads to blurred boundaries and the loss of small targets.

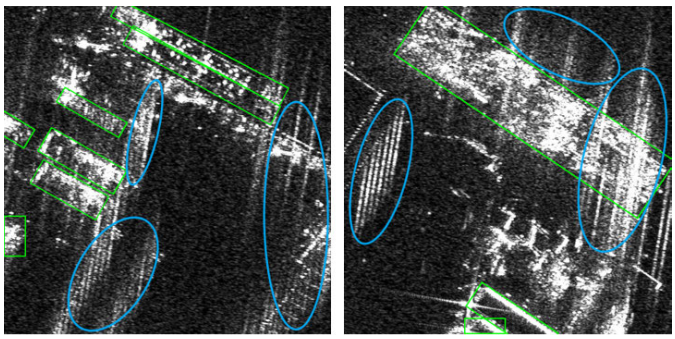


Fig. 1. Ship targets in complex scenes of SAR images (from the HRSID dataset). Green bounding boxes denote the annotated ship targets, while blue circles mark the inherent speckle noise.

Recently, there has been a paradigm shift toward explainable and physics-aware AI for SAR [4]. Aligning with this, we design a mechanism that leverages physical scattering differences to distinguish targets from coherent interference. To incorporate global context, recent advances explore frequency-domain features in deep networks [5], [6], [7].

However, the aforementioned methods struggle to precisely separate high and low-frequency information, making it difficult to accurately decouple high-frequency target details from random speckle noise. Thus, the wavelet transform (WT), possessing time–frequency localization characteristics, has become a research focus. Wang et al. [8] integrated Haar WT into a dual-backbone network to extract multifrequency texture features, enhancing the model's sensitivity to texture differences. Finder et al. [9] proposed a wavelet convolution that uses WT to decompose features into low- and high-frequency bands, capturing a global receptive field through low-frequency cascades while avoiding the quadratic computational overhead of large kernels. Although theoretically adaptable to the multiscale characteristics of SAR targets, this method has a critical flaw: it employs a simple summation strategy to reconstruct features, lacking a mech-

Received 1 March 2026; revised 8 May 2026; accepted 12 June 2026. Date of publication 16 June 2026; date of current version 1 July 2026. This work was supported in part by the Development Fund Project of Hebei Key Laboratory of Intelligent Information Perception and Processing under Grant SXX22138X002. *(Corresponding author: Weidi Xu.)*

The authors are with The 54th Research Institute of CETC, Shijiazhuang 050081, China, and also with Hebei Key Laboratory of Intelligent Information Perception and Processing, Shijiazhuang 050081, China (e-mail: haoshengsong1999@163.com; weidixu@126.com).

Digital Object Identifier 10.1109/LGRS.2026.3703971

1558-0571 © 2026 IEEE. All rights reserved, including rights for text and data mining, and training of artificial intelligence and similar technologies. Personal use is permitted, but republication/redistribution requires IEEE permission.
See https://www.ieee.org/publications/rights/index.html for more information.

Authorized licensed use limited to: Patuakhali Science and Technology Univ. Downloaded on October 15,2026 at 18:44:38 UTC from IEEE Xplore. Restrictions apply.

anism to distinguish "signal" from "noise" within high-frequency subbands. Consequently, unsuppressed speckle noise in these subbands (LH, HL, and HH) continuously accumulates and amplifies across network layers, ultimately destroying the semantic features required for precise detection.

To address this, we propose SAFANet, an end-to-end rotated object detection network featuring a novel SAR adaptive frequency attention mechanism SAFA-WT. SAFANet formulates feature extraction as a joint signal reconstruction and noise suppression task. In particular, it integrates a learnable ST module to physically suppress high-frequency clutter and an FA module to dynamically weight subbands based on global context. This design yields a dual benefit: cascaded low-frequency components efficiently capture global context, while adaptively processed high-frequency components filter speckle noise and preserve sharp target boundaries.

The main contributions of this letter can be summarized as follows.

1) We propose SAFA-WT, a novel wavelet-domain attention mechanism for SAR images. It integrates learnable ST and FA to efficiently decouple target details from speckle noise.
2) We develop the SAFANet architecture by embedding the SAFA-WT operator, significantly enhancing the network's capacity for global context modeling and clutter suppression.
3) Experiments on the SAR ship detection dataset (SSDD) and high-resolution SAR images dataset (HRSID) benchmark datasets validate SAFANet's superior detection performance, particularly demonstrating strong robustness in complex, heavily interfered scenes.

II. Methodology

This section details the SAFANet architecture, introduces the convolutional formulation of the WT, and mathematically formulates the proposed SAFA-WT mechanism.

A. Overall Architecture of SAFANet

As shown in Fig. 2, SAFANet consists of three main components: a feature extraction backbone, a multiscale feature fusion neck, and a multitask detection head.

The backbone employs an improved CSPDarknet. While standard 3×3 convolutions extract localized spatial features, their restricted effective receptive fields limit their robustness against speckle noise and efficiency in modeling global context. Thus, we replace the standard bottleneck layer with the SAFA-WT module.

The neck adopts a PAN-FPN variant to enhance multiscale target detection. It upsamples semantic-rich deep feature maps to align and concatenate with detail-rich shallow feature maps. Notably, the SAFA-WT module is reintroduced after this fusion to further refine and denoise the features.

Finally, each fused feature map connects to an independent detection head containing three parallel subbranches for predicting class, bounding box regression, and angle. Thus, SAFANet retains high-resolution processing in shallow layers while achieving a global receptive field and frequency-domain denoising via the SAFA-WT mechanism.

B. Construction of 2-D Haar WT/IWT Based on Convolution and Transposed Convolution

To losslessly extract and reconstruct frequency-domain features within CNNs, we transform the Discrete WT (DWT) and its inverse WT (IWT) into differentiable convolutions. This forms the foundation for frequency decomposition and recovery in the SAFA-WT module. We select the Haar wavelet for its computational efficiency and sensitivity to edge textures. Following [9], the 2-D Haar WT is executed using stride-2 depthwise convolutions with four filters:

$$f_{LL} = \frac{1}{2}\begin{bmatrix} 1 & 1 \\ 1 & 1 \end{bmatrix}, \quad f_{LH} = \frac{1}{2}\begin{bmatrix} 1 & -1 \\ 1 & -1 \end{bmatrix}$$
$$f_{HL} = \frac{1}{2}\begin{bmatrix} 1 & 1 \\ -1 & -1 \end{bmatrix}, \quad f_{HH} = \frac{1}{2}\begin{bmatrix} 1 & -1 \\ -1 & 1 \end{bmatrix} \tag{1}$$

where f_{LL} is the low-pass filter and f_{LH}, f_{HL}, and f_{HH} are a set of high-pass filters. For each input channel, the output of the convolution is

$$[X_{LL}, X_{LH}, X_{HL}, X_{HH}] = \text{Conv}\left(\left[f_{LL}, f_{LH}, f_{HL}, f_{HH}\right], X\right) \tag{2}$$

where X_{LL} is the low-frequency component of X and X_{LH}, X_{HL}, and X_{HH} are its horizontal, vertical, and diagonal high-frequency components, respectively.

The convolutional filters in (1) are strictly fixed to the standard Haar orthonormal basis and frozen during training. Thus, this linear operation is mathematically identical to the traditional DWT, ensuring the complete preservation of exact frequency decomposition and lossless reconstruction characteristics.

Conversely, applying the IWT can reconstruct the original signal, a process that can be implemented through transposed convolution

$$X = \text{Conv-transposed}\left(\left[f_{LL}, f_{LH}, f_{HL}, f_{HH}\right] [X_{LL}, X_{LH}, X_{HL}, X_{HH}]\right). \tag{3}$$

C. SAFA-WT: SAR Adaptive Frequency Attention Mechanism

As the core component, SAFA-WT is embedded into the backbone and neck networks (Fig. 2) to resolve noise interference during multiscale feature extraction. Previous cascaded wavelet methods expanded the receptive field but used simple summation for reconstruction, inadvertently aggregating high-frequency speckle noise in SAR images. To address this, we propose the SAFA-WT attention mechanism, introducing a two-stage "Filter-and-Refine" strategy—forward decomposition and adaptive reconstruction—in the wavelet domain.

In the forward decomposition, given an input tensor X, a multiscale wavelet decomposition is first executed. Let $X_{LL}^{(i-1)}$ be the input of the ith level ($i = 1, \ldots, L$), the decomposition and convolution transformations are defined as follows:

$$X_{LL}^{(i)}, X_{LH}^{(i)}, X_{HL}^{(i)}, X_{HH}^{(i)} = \text{DWT}\left(X_{LL}^{(i-1)}\right) \tag{4}$$

$$Y_{band}^{(i)} = \text{Conv}_{3\times 3}\left(X_{band}^{(i)}\right), \text{band} \in \{LL, LH, HL, HH\} \tag{5}$$

where $\text{Conv}_{3\times 3}$ denotes depthwise convolution used to extract features within each frequency band.

The adaptive reconstruction proceeds inversely from the deepest level L to level 0, introducing two core components here: learnable ST and FA.

Authorized licensed use limited to: Patuakhali Science and Technology Univ. Downloaded on October 15,2026 at 18:44:38 UTC from IEEE Xplore. Restrictions apply.

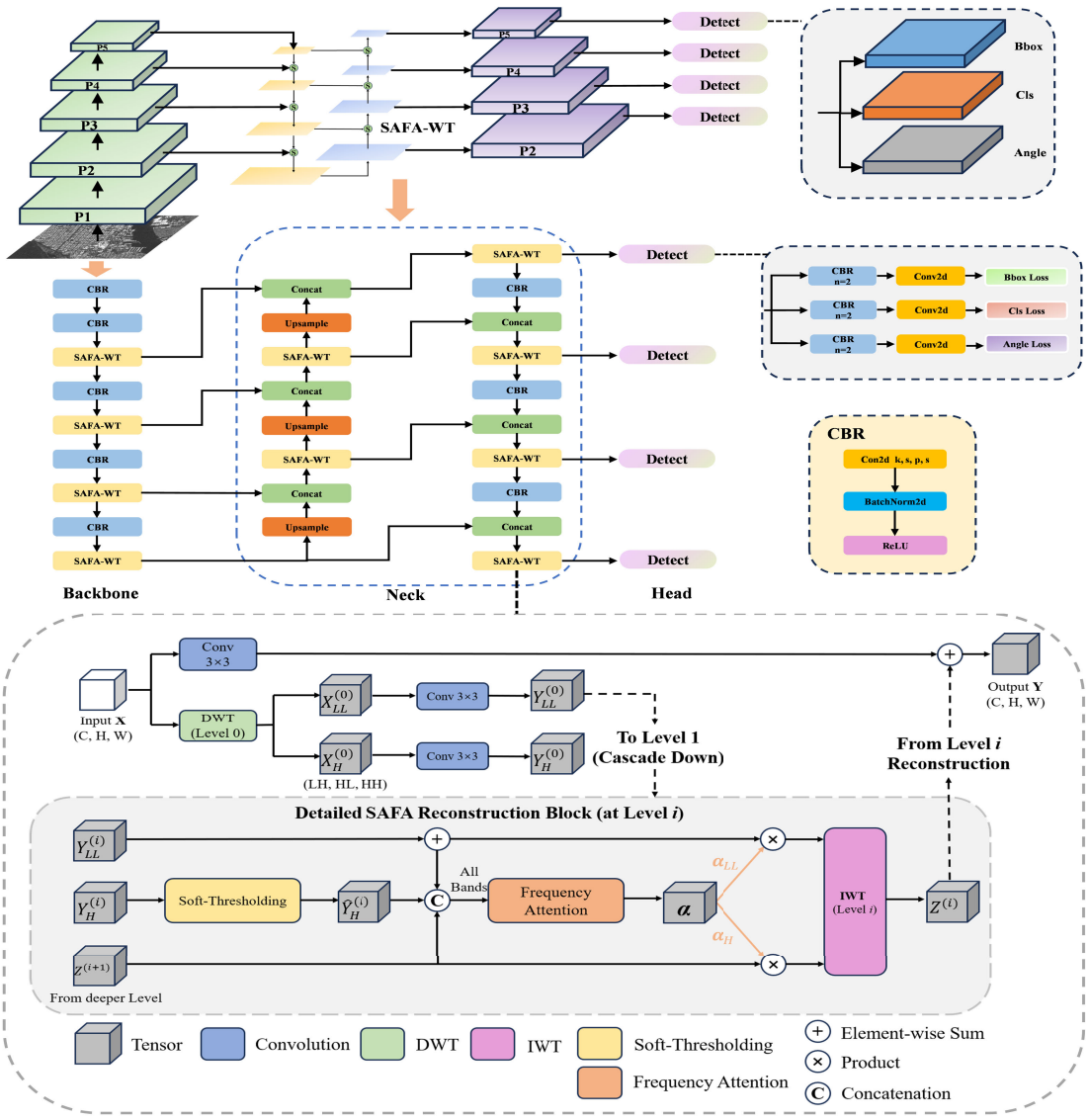


Fig. 2. Framework of the SAFANet model.

1) Learnable ST (Physical Denoising): SAR speckle noise is predominantly composed of high-frequency random components. Inspired by wavelet denoising theory, we apply an ST operation to the high-frequency components. Unlike traditional fixed-threshold methods, a learnable parameter $\tau \in \mathbb{R}^C$ is introduced for each channel

$$\hat{Y}_k^{(i)} = S_\tau\left(Y_k^{(i)}\right) = \text{sgn}\left(Y_k^{(i)}\right) \cdot ReLU\left(|Y_k^{(i)}| - \tau\right) \tag{6}$$

where $k \in \{LH, HL, HH\}$. This operation physically suppresses low-amplitude clutter (noise) while retaining strong edge responses (targets).

2) FA (Adaptive Weighting): To quantify the importance of each frequency band after denoising, we design a lightweight FA module. This module concatenates the low-frequency component, enhanced by the deep contextual information $Z^{(i+1)}$, with the denoised high-frequency components

$$F_{cat}^{(i)} = \text{Concat}\left(\left[Y_{LL}^{(i)} + Z^{(i+1)}, \hat{Y}_{LH}^{(i)}, \hat{Y}_{HL}^{(i)}, \hat{Y}_{HH}^{(i)}\right]\right). \tag{7}$$

Through global average pooling (GAP) and a fully convolutional network (FCN), the attention vector $\alpha \in \mathbb{R}^{4C}$ is generated

$$\alpha = \sigma\left(W_2\delta\left(W_1\text{GAP}\left(F_{cat}^{(i)}\right)\right)\right) \tag{8}$$

where σ is the Sigmoid function and δ is the ReLU function. The attention weights are split into $\alpha_{LL}, \alpha_{LH}, \alpha_{HL}, \alpha_{HH}$, corresponding to the weight coefficients of each respective frequency band.

Subsequently, a weighted reconstruction is performed. The final output $Z^{(i)}$ of the ith level is obtained by applying the inverse discrete WT to reconstruct the weighted components

$$Z^{(i)} = \text{IWT}\left(\alpha_{LL} \odot \left(Y_{LL}^{(i)} + Z^{(i+1)}\right), \sum_{k\in\{High\}} \alpha_k \odot \hat{Y}_k^{(i)}\right) \tag{9}$$

where $\odot$ denotes elementwise multiplication. This design enables the network to dynamically suppress noise through parameters τ and α_k and enhance structural features through α_{LL}.

Authorized licensed use limited to: Patuakhali Science and Technology Univ. Downloaded on October 15,2026 at 18:44:38 UTC from IEEE Xplore. Restrictions apply.

TABLE I
RESULTS OF COMPARATIVE EXPERIMENTS

Module	SSDD				HRSID			
	P (%)	R (%)	mAP_{50} (%)	mAP (%)	P (%)	R (%)	mAP_{50} (%)	mAP (%)
R-RetinaNet [2]	76.18	76.53	70.83	38.00	72.14	79.20	69.10	37.07
R-FCOS [12]	92.30	97.18	93.70	52.22	82.56	87.25	82.87	46.20
R-Faster CNN [1]	85.64	88.12	84.65	43.14	84.53	72.78	82.34	45.65
ReDet [13]	89.36	92.09	88.65	54.83	77.93	83.11	80.12	44.34
R3Det [14]	85.22	86.00	82.72	46.40	82.66	86.14	83.70	46.96
ELLK-Net [15]	90.30	92.50	89.70	-	89.60	87.70	79.70	-
GWD [16]	94.20	95.40	90.00	52.50	88.20	81.70	91.50	66.00
YOLOv8-OBB [17]	94.77	95.39	98.09	70.41	90.82	83.88	92.83	67.30
YOLOv11-OBB [17]	96.95	95.61	98.60	69.86	91.21	84.57	93.10	67.65
YOLOv12-OBB [18]	97.66	96.62	99.04	70.78	90.95	82.25	91.66	63.91
YOLO26-OBB [17]	95.84	96.43	98.86	71.12	91.16	82.91	92.68	67.82
Ours	**97.73**	**97.67**	**99.30**	**71.97**	**93.89**	**87.78**	**94.91**	**69.33**

III. EXPERIMENTS

A. Datasets

To evaluate SAFANet's denoising and multiscale extraction capabilities in complex scenes, we utilize the SSDD [10] and HRSID [11] datasets. SSDD comprises 1160 images (1–15-m resolution) showcasing multiscale ships under diverse environmental clutter. Following the official protocol, images ending with digits 1 and 9 are allocated for testing, and the rest for training. HRSID contains 5604 high-resolution (0.5–3 m) images characterized by intense speckle noise, varying sea states, and severe inshore port interference. We adopt its default configuration of 3642 training and 454 testing images. The speckle interference and scale variations in these datasets perfectly match our core research objectives, providing a rigorous testbed for the proposed method.

B. Evaluation Metrics and Experimental Details

We utilize precision (P), recall (R), and mean average precision (mAP) as evaluation metrics.

All experiments in this study were conducted using the PyTorch 1.13.0 framework, with AdamW as the optimizer. The models were trained with a batch size of 32 and an input image size of 640, for 300 epochs, with a learning rate of 0.001 and a momentum of 0.9. The intersection over union (IoU) thresholds for detection evaluation were set to 0.5 and 0.50:0.95, and no pretrained weights were used.

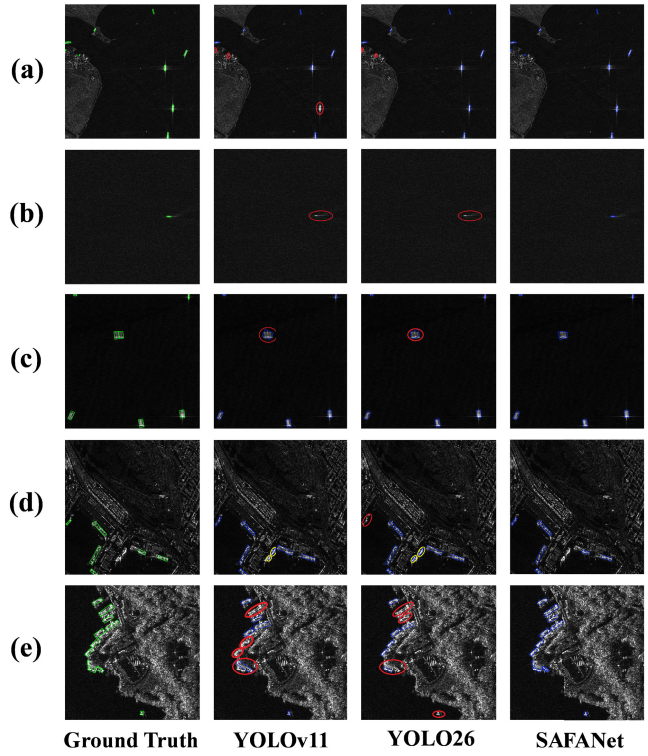


Fig. 3. Visual comparison of SAFANet detection results. The yellow circles indicate false alarm targets, and the red circles represent missed detection targets. (a)–(c) Offshore scenarios and (d)–(e) inshore scenarios.

C. Comparative Experiments

To evaluate the proposed method, we conducted comprehensive comparative experiments on the SSDD and HRSID datasets. Table I summarizes the comparison results between our method and several current mainstream rotated object detection algorithms, covering both two-stage detection methods and advanced single-stage detection methods.

As shown in Table I, SAFANet achieves optimal performance on both datasets. On SSDD, it reaches 99.30% mAP_{50} and 71.97% mAP, outperforming the second-best YOLOv12-OBB by 0.26% and 1.19%, respectively. On the more challenging HRSID dataset, its advantages are more pronounced, achieving a peak 94.91% mAP_{50} and 69.33% mAP, which surpasses YOLOv11-OBB by 1.81% and 1.68%, respectively. Furthermore, SAFANet consistently outperforms the recently released YOLO26-OBB, proving its ability to suppress speckle-induced false alarms while maintaining high sensitivity for small targets.

To qualitatively evaluate the proposed method, Fig. 3 visualizes the detection results of the ground truth annotations, two mainstream comparative models (YOLOv11, YOLO26), and SAFANet in various typical SAR scenes. The red and yellow circles in the figure denote missed detections and false alarms, respectively.

Authorized licensed use limited to: Patuakhali Science and Technology Univ. Downloaded on October 15,2026 at 18:44:38 UTC from IEEE Xplore. Restrictions apply.

TABLE II
RESULTS OF SAFA-WT ABLATION EXPERIMENTS

ST	FA	$Precision(\%)$	$Recall(\%)$	mAP_{50} (%)	mAP (%)
×	×	89.49	79.93	89.65	57.67
×	✓	90.54	84.08	92.14	63.68
✓	×	91.52	83.74	92.10	62.70
✓	✓	**93.89**	**87.78**	**94.91**	**69.33**

In open-sea scenes [Rows 3(a)–(c)], YOLOv11 and YOLO26 struggle with small, low-signal targets [Row 3(b)] and misclassify closely situated ships as single objects [Row 3(c)]. Conversely, SAFANet's enhanced feature representation accurately identifies these weak targets. In complex inshore scenes [Rows 3(d) and (e)], strong scattering from port facilities causes the YOLO models to suffer severe missed detections and false alarms, often misjudging land structures as ships. SAFANet, however, effectively suppresses this high-frequency clutter while completely preserving target structures, demonstrating superior robustness against speckle noise and complex backgrounds.

D. Ablation Studies

In this section, we conducted extensive ablation experiments on the HRSID dataset to verify the effectiveness of the proposed modules. Keeping the overall architecture unchanged, we replace the proposed SAFA-WT with WTConv [9] as the baseline to isolate the individual contributions of the WT structure, ST, and FA. The results are summarized in Table II.

As shown in Table II, the baseline model's limited accuracy confirms that standard multifrequency aggregation introduces speckle noise. Integrating the FA module boosts recall from 79.93% to 84.08%, indicating that adaptive weighting helps focus on target-dominated subbands to recover missed targets. Independently, the ST module improves precision by 2.03% over the baseline, verifying its physical denoising role in suppressing high-frequency clutter and reducing false alarms. Finally, the combined SAFA-WT structure achieves optimal performance, boosting *mAP* to 69.33% (an 11.66 improvement). This demonstrates their complementarity: ST provides clean features, while FA optimizes representation, ensuring precise detection in complex backgrounds.

IV. CONCLUSION

To address speckle noise and multiscale target variations in SAR ship detection, this letter proposes SAFANet. Its core, the SAFA-WT mechanism, integrates learnable ST and FA within the wavelet domain. This design adaptively suppresses high-frequency clutter and dynamically weights multifrequency features, effectively decoupling target details from speckle noise. Experiments on the SSDD and HRSID datasets demonstrate SAFANet's superiority in strongly interfered scenes. Notably, on the challenging HRSID dataset, it achieves a 2.23% mAP_{50} improvement over YOLO26-OBB, substantially reducing false alarms and missed detections. Future work will extend this wavelet-domain concept to optical-SAR cross-modal domain adaptation.

REFERENCES

[1] S. Ren, K. He, R. Girshick, and J. Sun, "Faster R-CNN: Towards real-time object detection with region proposal networks," *IEEE Trans. Pattern Anal. Mach. Intell.*, vol. 39, no. 6, pp. 1137–1149, Jun. 2017.
[2] T.-Y. Lin, P. Goyal, R. Girshick, K. He, and P. Dollár, "Focal loss for dense object detection," in *Proc. IEEE Int. Conf. Comput. Vis. (ICCV)*, Oct. 2017, pp. 2999–3007.
[3] J. Redmon, S. Divvala, R. Girshick, and A. Farhadi, "You only look once: Unified, real-time object detection," in *Proc. IEEE Conf. Comput. Vis. Pattern Recognit. (CVPR)*, May 2016, pp. 779–788.
[4] M. Datcu, Z. Huang, A. Anghel, J. Zhao, and R. Cacoveanu, "Explainable, physics-aware, trustworthy artificial intelligence: A paradigm shift for synthetic aperture radar," *IEEE Geosci. Remote Sens. Mag.*, vol. 11, no. 1, pp. 8–25, Mar. 2023.
[5] Z. Qin, P. Zhang, F. Wu, and X. Li, "FcaNet: Frequency channel attention networks," in *Proc. IEEE/CVF Int. Conf. Comput. Vis. (ICCV)*, Oct. 2021, pp. 763–772.
[6] H. Song, W. Xu, L. Wang, J. Chen, and H. Yu, "SFONet: A novel joint spatial-frequency domain algorithm for multiclass ship oriented detection in SAR images," *IEEE J. Sel. Topics Appl. Earth Observ. Remote Sens.*, vol. 18, pp. 20262–20278, 2025.
[7] D. Li, Q. Liang, H. Liu, Q. Liu, H. Liu, and G. Liao, "A novel multidimensional domain deep learning network for SAR ship detection," *IEEE Trans. Geosci. Remote Sens.*, vol. 60, 2022, Art. no. 5203213, doi: 10.1109/TGRS.2021.3062038.
[8] S. Wang, Z. Cai, and J. Yuan, "Automatic SAR ship detection based on multifeature fusion network in spatial and frequency domains," *IEEE Trans. Geosci. Remote Sens.*, vol. 61, 2023, Art. no. 4102111, doi: 10.1109/TGRS.2023.3267495.
[9] S. E. Finder, R. Amoyal, E. Treister, and O. Freifeld, "Wavelet convolutions for large receptive fields," in *Proc. Eur. Conf. Comput. Vis. (ECCV)*, 2024, pp. 363–380.
[10] T. Zhang et al., "SAR ship detection dataset (SSDD): Official release and comprehensive data analysis," *Remote Sens.*, vol. 13, no. 18, p. 3690, Sep. 2021.
[11] S. Wei, X. Zeng, Q. Qu, M. Wang, H. Su, and J. Shi, "HRSID: A high-resolution SAR images dataset for ship detection and instance segmentation," *IEEE Access*, vol. 8, pp. 120234–120254, 2020.
[12] H. Law and J. Deng, "CornerNet: Detecting objects as paired keypoints," *Int. J. Comput. Vis.*, vol. 128, no. 3, pp. 642–656, Mar. 2020.
[13] J. Han, J. Ding, N. Xue, and G.-S. Xia, "ReDet: A rotation-equivariant detector for aerial object detection," in *Proc. IEEE/CVF Conf. Comput. Vis. Pattern Recognit. (CVPR)*, Jun. 2021, pp. 2785–2794.
[14] X. Yang, J. Yan, Z. Feng, and T. He, "R3Det: Refined single-stage detector with feature refinement for rotating object," in *Proc. AAAI Conf. Artif. Intell.*, May 2021, pp. 3163–3171.
[15] J. Shen, L. Bai, Y. Zhang, M. Chowdhuray Momi, S. Quan, and Z. Ye, "ELLK-Net: An efficient lightweight large kernel network for SAR ship detection," *IEEE Trans. Geosci. Remote Sens.*, vol. 62, 2024, Art. no. 5221514, doi: 10.1109/TGRS.2024.3451399.
[16] X. Yang, J. Yan, Q. Ming, W. Wang, X. Zhang, and Q. Tian, "Rethinking rotated object detection with Gaussian Wasserstein distance loss," in *Proc. Int. Conf. Mach. Learn.*, 2021, pp. 11830–11841.
[17] G. Jocher. (2026). *Ultralytics*. [Online]. Available: https://github.com/ultralytics/ultralytics
[18] Y. Tian, Q. Ye, and D. Doermann, "YOLOv12: Attention-centric real-time object detectors," 2025, *arXiv:2502.12524*.

Authorized licensed use limited to: Patuakhali Science and Technology Univ. Downloaded on October 15,2026 at 18:44:38 UTC from IEEE Xplore. Restrictions apply.